CAP 6908 DIS (Fall 2026)

When does a LiDAR detector stop relying on the object?

A stratified study of evidence dependence in 3D detection confidence

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Week of Oct 12. Milestone: midpoint report. Baseline stratification plus stratified counterfactual results for the primary detector, with figures and a written interpretation.

Progress

No new forwards were run, and the checkpoint, the bins, and the metrics are unchanged. The weeks 6–7 update sent with this PDF is the pilot and the full stratified run. This PDF is the midpoint report named for this week. It puts the week-4 baseline in the same document as those counterfactual results, with the figures and a written interpretation. The deletion rates in the report are the full 11,125 cars, the same rates as the weeks 6–7 note. The CVPR-format draft is a separate document and is not this update.

Next steps

Week of Oct 19 repeats the same protocol on a one-stage detector and a two-stage detector. No second checkpoint has been run.

Attached below

The midpoint report starts on the next page.

Midpoint report

Baseline stratification and stratified counterfactual results

Public PointPillars checkpoint, KITTI validation split, Car

Claim

Hold one public KITTI LiDAR checkpoint fixed. Measure how much its confidence depends on points inside the predicted box versus the surrounding scene, and whether that dependence is stronger when the object's own evidence is weak (far, sparse, or small). OccAM already showed residual confidence after emptying the box. The contribution is the stratified measurement, not a new detector or a new attribution method.

Setup

Three populations are used below, and they are not one denominator.

Official average precision is computed on ground-truth cars. A detection counts at the official Car 3D overlap of 0.7. The number quoted is moderate 3D AP at 11 recall positions. AP at 40 recall positions is stored with the run and is not the number compared with the published 77.28.

Confidence and expected calibration error are computed on week-3 Car predictions kept at score at least 0.1. A prediction counts as correct when its stored 3D overlap with a ground-truth car is at least 0.7. ECE uses 10 equal-width score bins, empty bins omitted. Accuracy in a bin is the fraction of those predictions at that overlap.

The deletion sample is a third set: week-3 Car predictions with score at least 0.7 and at least one interior point. That is 11,125 cars. Of those, 4,864 are within 20 m, 4,868 are between 20 and 40 m, and 1,393 are at 40 m or farther. By interior points, 370 have fewer than 20, 3,419 have 20 to 99, and 7,336 have 100 or more. By box volume, 1,299 are under $8\,\mathrm{m}^3$, 9,082 are between 8 and $12\,\mathrm{m}^3$, and 744 are $12\,\mathrm{m}^3$ or larger. Ground-truth match was recorded on 10,478 of the 11,125 and was not required. The week-3 match flag is an overlap of 0.5. It is not the AP bar, and it is not an entry rule for this sample.

The bins are unchanged. Range is $[0, 20)$, $[20, 40)$, and $[40, 80]$ with overflow in the last bin. Interior points are $[0, 20)$, $[20, 100)$, and $[100, +\infty)$. Volume is $[0, 8)$, $[8, 12)$, and $[12, +\infty)$ cubic meters.

The detector is the public PointPillars checkpoint `pointpillar_7728.pth`. Fractions are 0, 0.25, 0.5, 0.75, and 1, with seed 6908, and one target box per forward. Interior removal deletes points inside the predicted box. The placebo deletes points in a same-size box placed in an empty region. Context removal deletes points outside the box. Fraction 0 deletes nothing. Placebo placement was available for all 11,125 targets.

A returned box is the same detection when its 3D overlap with the original predicted box is at least 0.5. Retained confidence is the new score divided by the original score when that bar is met, and 0 when it is not. A miss stays in the mean. Score survival also requires the returned score to be at least 0.5. IoU survival is the overlap bar alone. A high score on a different car is not retained confidence.

The definitions were checked on a pilot of 200 cars before the full split: 67 within 20 m, 67 between 20 and 40 m, and 66 at 40 m or farther. Both metrics fell at fraction 1 in every range bin, so the

decision was GO. The rates below are the full 11,125, not that pilot.

Baseline

Unstratified Car 3D AP at 11 recall positions is 86.4659 on easy, 77.2898 on moderate, and 74.6573 on hard. The number used from here is moderate **77.2898**.

Table 1 is that moderate number by bin. The two n columns count different sets. Average precision falls from 87.8298 within 20 m, to 72.1397 between 20 and 40 m, and to 14.0610 at 40 m and farther. Those ground-truth bins hold 5,274, 5,858, and 3,253 cars. By interior points it falls from 87.5086 when a labeled car has at least 100 points, to 49.5029 with 20 to 99 points, and to 2.2102 with fewer than 20. Volume does not follow that pattern. Cars under 8 m^3 are at 69.1815, cars between 8 and 12 m^3 are at 76.8579, and cars of 12 m^3 and larger are at 29.0127. Those edges were not retuned.

ECE on all Car predictions is **0.1488**. It is 0.0636 within 20 m, 0.1404 between 20 and 40 m, and 0.2039 beyond 40 m. It is 0.2004 below 20 interior points, 0.1841 from 20 to 99 points, and 0.0451 at 100 points or more. By volume the spread is smaller: 0.1310 under 8 m^3, 0.1475 between 8 and 12 m^3, and 0.1815 at 12 m^3 and larger. Range and point count already show the known pattern: AP falls and ECE rises where the object's own evidence is weak. Size does not.

Table 1: Baseline by stratum. Moderate AP is official 3D AP at 11 recall positions on ground-truth cars, overlap 0.7. Mean confidence is rounded to three decimals. ECE is the week-4 value. Predictions are boxes kept at score 0.1, matched at 3D overlap 0.7. The two n columns are different populations.

Bin	GT n	AP	Pred. n	Conf.	ECE
All	14385	77.2898	33133	0.472	0.1488
0–20 m	5274	87.8298	6987	0.729	0.0636
20–40 m	5858	72.1397	12109	0.528	0.1404
40 m and farther	3253	14.0610	14037	0.297	0.2039
0–19 points	2161	2.2102	13506	0.242	0.2004
20–99 points	4457	49.5029	9922	0.495	0.1841
100 and above	7767	87.5086	9705	0.770	0.0451
Under 8 m^3	2358	69.1815	2536	0.615	0.1310
8–12 m^3	10781	76.8579	27590	0.467	0.1475
12 m^3 and larger	1246	29.0127	3007	0.399	0.1815

Figure 1 is the reliability diagram for range. The horizontal axis is mean confidence in a score bin, and the vertical axis is the fraction of those predictions whose 3D overlap with a ground-truth car is at least 0.7. A curve under the diagonal is a score that is too high for how often the box is right. Figure 2 is the same diagram for interior point count and for volume. Figure 3 is where the scores sit. The vertical axis is the number of predictions.

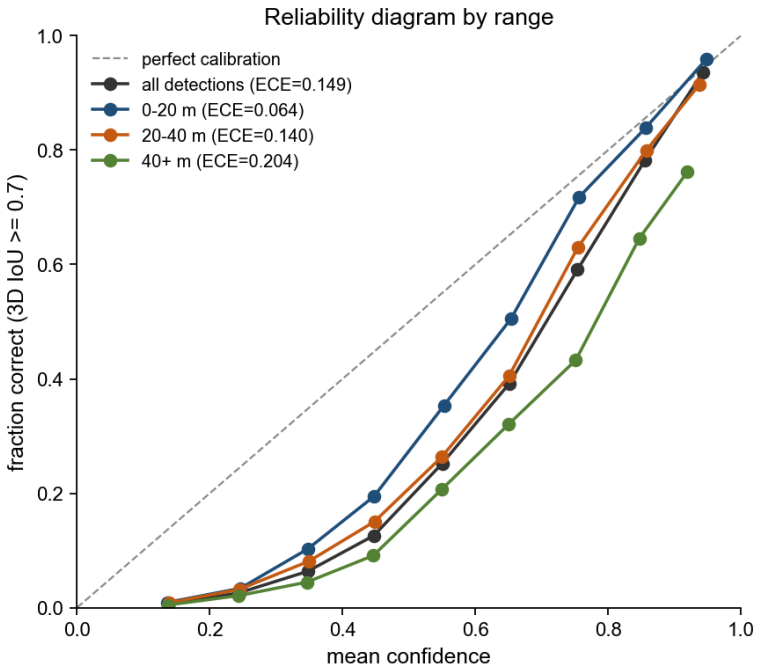


Figure 1: Reliability by sensor range. Predictions at 0–20 m stay closest to the diagonal (ECE 0.0636). Predictions at 40 m and farther fall furthest below it (ECE 0.2039).

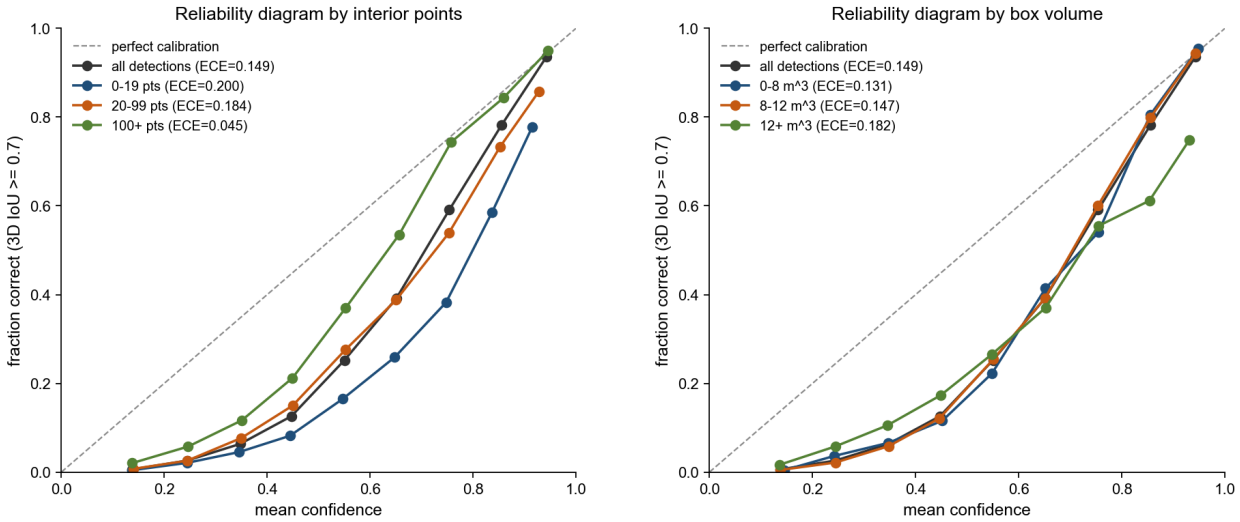


Figure 2: Reliability by interior point count (left) and by predicted box volume (right). Axes match Figure 1. Dense predictions stay closest to the diagonal. The size curves stay closer together than the range curves.

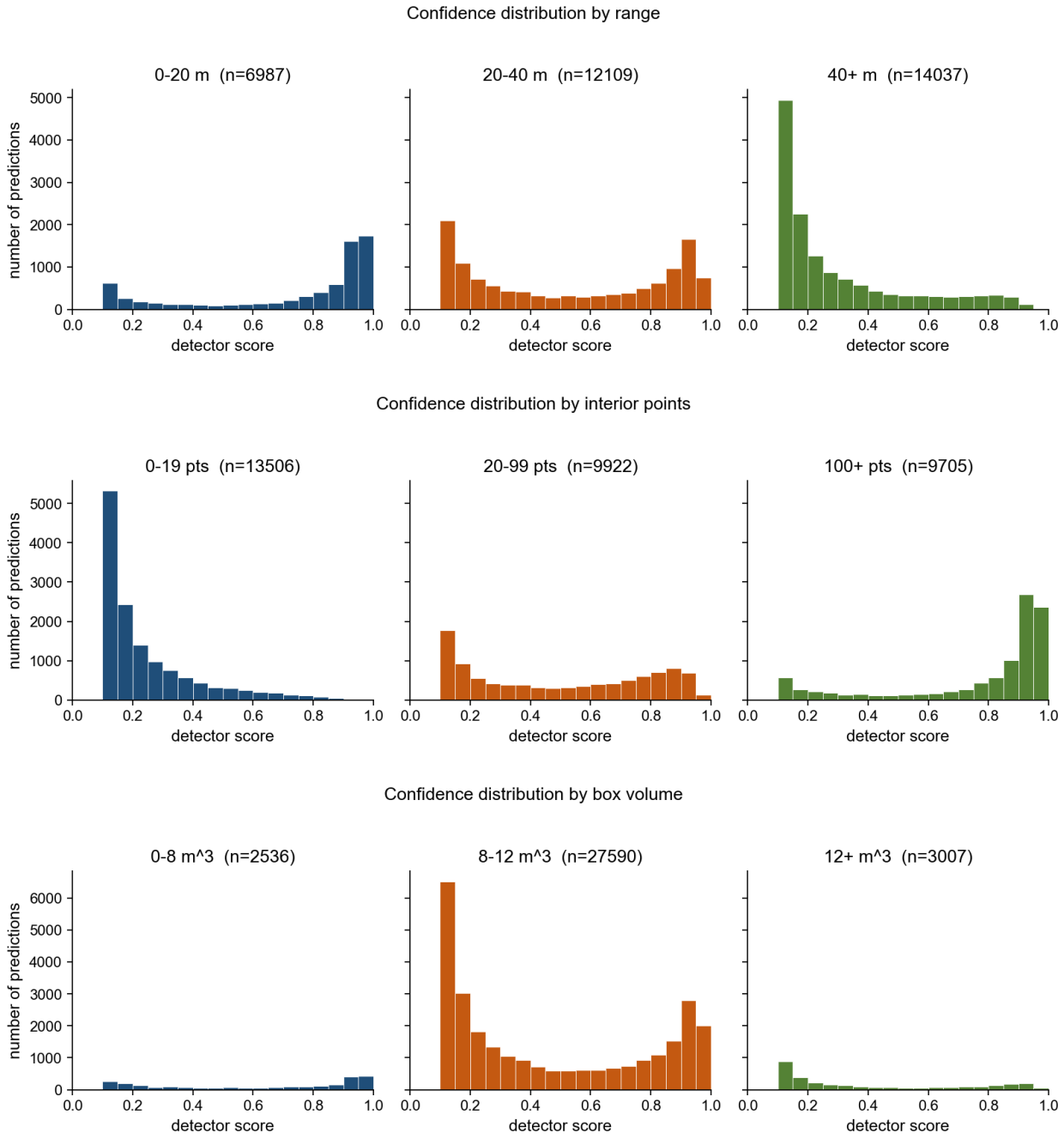


Figure 3: Confidence distributions by range, interior point count, and box volume. Each row is one split, and each row already contains its three bins. The vertical axis is the number of predictions. Near cars and cars with many interior points pile up near 1. Far cars and cars with few interior points pile up near the 0.1 output threshold.

Counterfactual results

Figure 4 is mean retained confidence against removal fraction, split by range, for the interior edit and the two controls. The interior curve stays near 1 through the middle fractions and then falls,

and the far panel falls first. At fraction 1, mean retained confidence is 0.193 within 20 m, 0.149 between 20 and 40 m, and 0.074 at 40 m and farther. Score survival is 658 of the 4,864 near cars, 503 of the 4,868 mid-range cars, and 53 of the 1,393 far cars. A returned box still meets the overlap bar for 2,213 near cars, 1,689 mid-range cars, and 252 far cars. A box comes back more often than a score of 0.5. Those rows are not detections that vanished. Table 2 is fraction 1 for all three conditions.

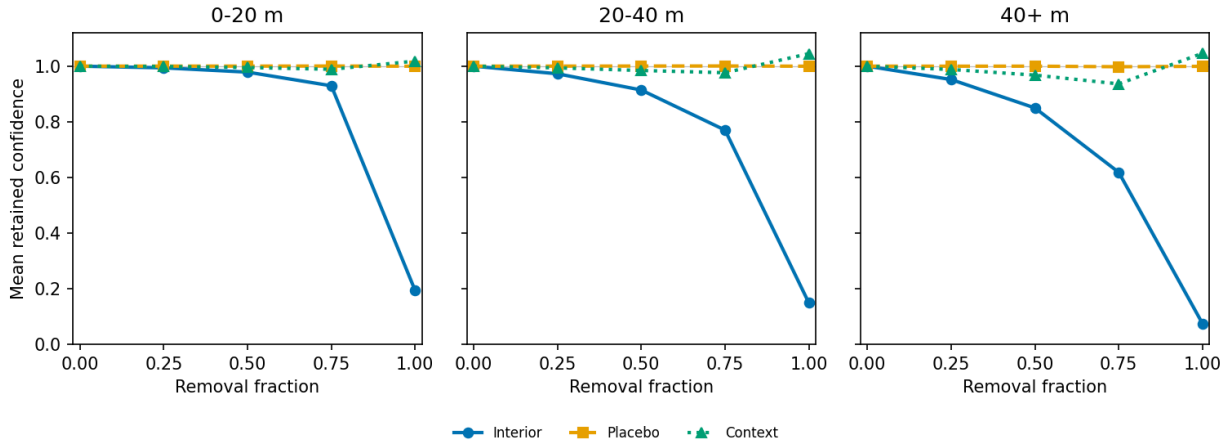


Figure 4: Mean retained confidence against removal fraction, one panel per range bin. Interior removal falls, and it falls sooner at longer range. The placebo and the context edit stay near 1. At fraction 1 the context curve is slightly above 1. A miss is stored as zero, so deaths stay in the mean.

Table 2: Fraction 1 on all 11,125 eligible cars, by range. Retained confidence is rounded to three decimals. The three range bins have $n = 4864$, 4868, and 1393.

Condition	Range	Mean removed	Mean retained	Score survival	IoU survival
Interior	0–20 m	1093	0.193	658/4864	2213/4864
Interior	20–40 m	127	0.149	503/4868	1689/4868
Interior	40+ m	33	0.074	53/1393	252/1393
Placebo	0–20 m	272	1.000	4864/4864	4864/4864
Placebo	20–40 m	22	1.000	4866/4868	4867/4868
Placebo	40+ m	5	0.999	1393/1393	1393/1393
Context	0–20 m	115970	1.018	4845/4864	4851/4864
Context	20–40 m	117733	1.045	4856/4868	4858/4868
Context	40+ m	118224	1.046	1389/1393	1391/1393

Figure 5 is the same three conditions split by interior point count. At fraction 1 the interior mean is 0.086 below 20 points, 0.098 from 20 to 99 points, and 0.190 at 100 points or more. The sparse bin is already at 0.519 by fraction 0.75. Score survival is 16 of 370 sparse cars, 197 of 3,419 in the middle bin, and 1,001 of 7,336 dense cars. Overlap survival in those same bins is 71, 839, and 3,244. Both metrics are down from fraction 0 in every point-count bin, and the drop arrives sooner when the box already holds fewer points.

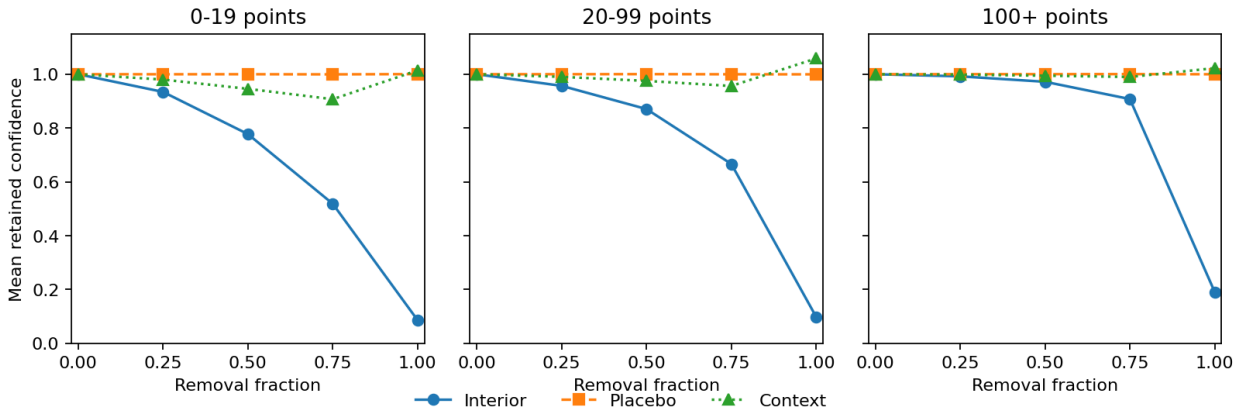


Figure 5: Mean retained confidence against removal fraction, one panel per interior-point bin. The sparse bin falls earlier than the dense bin. The placebo stays near 1. Context dips and then returns to about 1 at fraction 1, when the cloud is the object alone.

Figure 6 is the same split by box volume. At fraction 1 the interior mean is 0.177 under 8 m^3, 0.154 between 8 and 12 m^3, and 0.178 at 12 m^3 and larger. The three curves stay together across the fractions. Volume does not separate the result the way range and point count do. Table 3 is the fraction-1 interior cut for both of those axes.

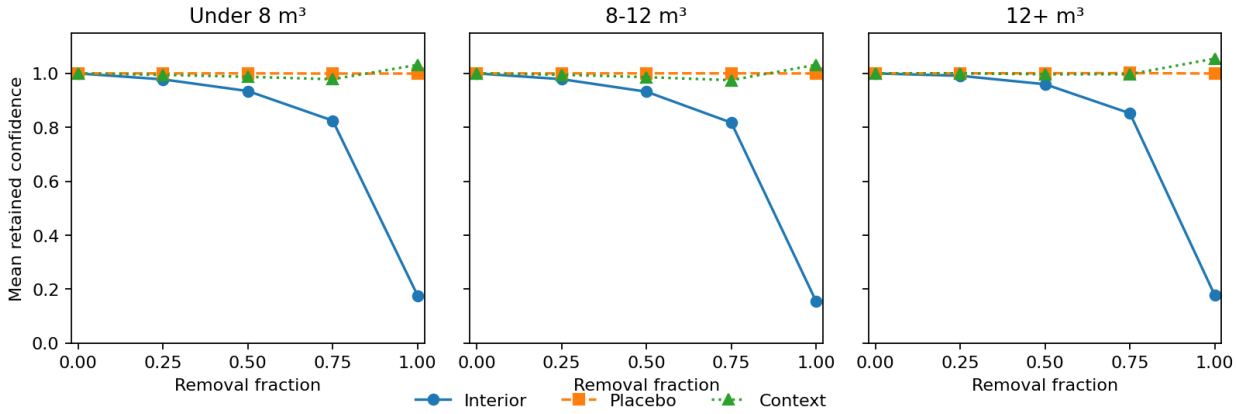


Figure 6: Mean retained confidence against removal fraction, one panel per box-volume bin. The interior curves fall together. Volume does not reproduce the range split or the point-count split.

Table 3: Interior removal at fraction 1, by interior point count and by box volume. Same retention rule as Table 2.

Bin	n	Retained	Score survival	IoU survival
0–19 points	370	0.086	16/370	71/370
20–99 points	3419	0.098	197/3419	839/3419
100 and above	7336	0.190	1001/7336	3244/7336
Under 8 m^3	1299	0.177	163/1299	536/1299
8–12 m^3	9082	0.154	948/9082	3340/9082
12 m^3 and larger	744	0.178	103/744	278/744

The joint cells, all 27 of them, are in `week7_table.csv`. They are not narrated here. The claim uses the marginal bins above.

Controls

The placebo does not reproduce the interior drop. At fraction 1 the placebo mean is 1.000 within 20 m, 1.000 between 20 and 40 m, and 0.999 beyond 40 m. The empty box usually contains fewer points than the car. The placebo removes about 272 points within 20 m, 22 points between 20 and 40 m, and 5 points beyond 40 m. The interior edit removes about 1,093 points within 20 m, 127 between 20 and 40 m, and 33 beyond 40 m. This rules out a random deletion of some same-sized patch of the scan. It does not rule out a matched point-count deletion.

Context removal does not reproduce the drop either. Deleting the surrounding scan and leaving the object raises mean retained confidence at fraction 1 to 1.018 within 20 m, 1.045 between 20 and 40 m, and 1.046 beyond 40 m. The curve does move before that. The far bin at fraction 0.75 is 0.936, and the object-alone cloud at fraction 1 comes back up, slightly above 1. The same pattern holds on the point-count and size margins in Figures 5 and 6: the placebo stays at 1, and context ends at or slightly above 1.

Figures 7–9 are examples from the same run, not the means in Table 2. Figures 7 and 9 are one near car. Figure 8 is one far car.

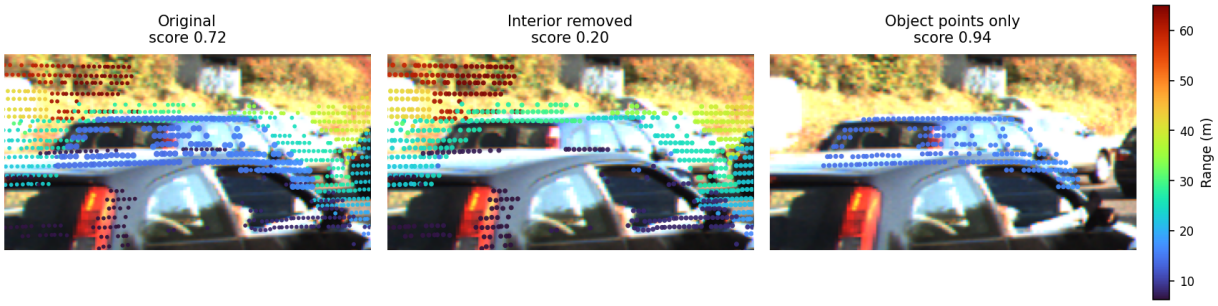


Figure 7: Frame 006853, prediction 5, 18 m from the sensor, 189 interior points, original score 0.72. Removing the interior points drops the stored score to 0.20. Keeping only those interior points, and deleting the rest of the scan, leaves a score of 0.94.

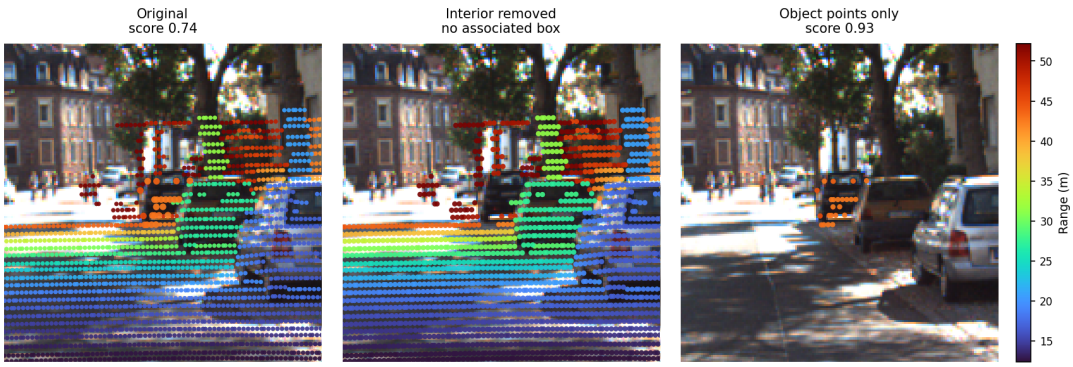


Figure 8: Frame 007169, prediction 6, in the 40+ m bin, 23 interior points, original score 0.74. After the interior points are removed, no new Car box meets the IoU bar. The same 23 points alone return a score of 0.93.

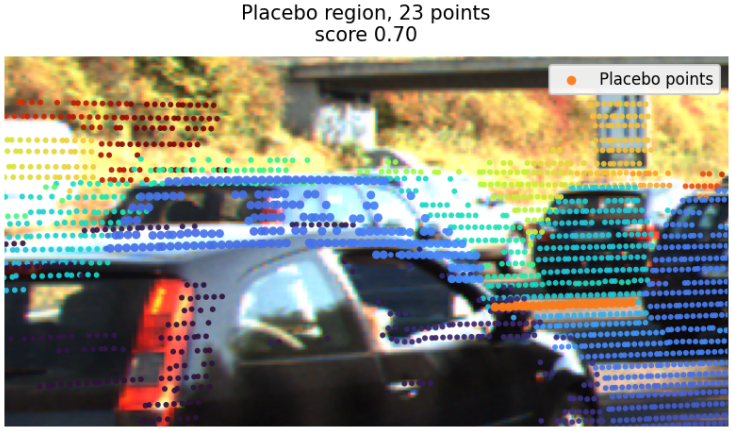


Figure 9: The same car as Figure 7. The marked points are the lidar returns inside the placebo region, 23 points in an empty box beside the car. Deleting them leaves the stored score at 0.70. Deleting the 189 points on the car is what moves the score.

Interpretation

The baseline and the deletion answer different questions. The baseline says detection and calibration are already worse far away and on sparse cars. Average precision falls and calibration error rises on those bins, and box volume does not follow that pattern. The deletion says confidence on the high-score cars falls when the interior points go away, and it falls sooner in those same weak bins.

Emptying the box moves both retained confidence and score survival down in every range bin and every point-count bin. Residual confidence at fraction 1 is not zero. That is the OccAM observation, broken out by these bins. The drop is steeper for far cars and for cars with fewer interior points. That is the part of the claim the interior run supports. Box volume does not show the same split.

IoU survival is higher than score survival at fraction 1. A box comes back more often than a score of 0.5.